Variable Length Subnet Mask

Mujahid Tabassum

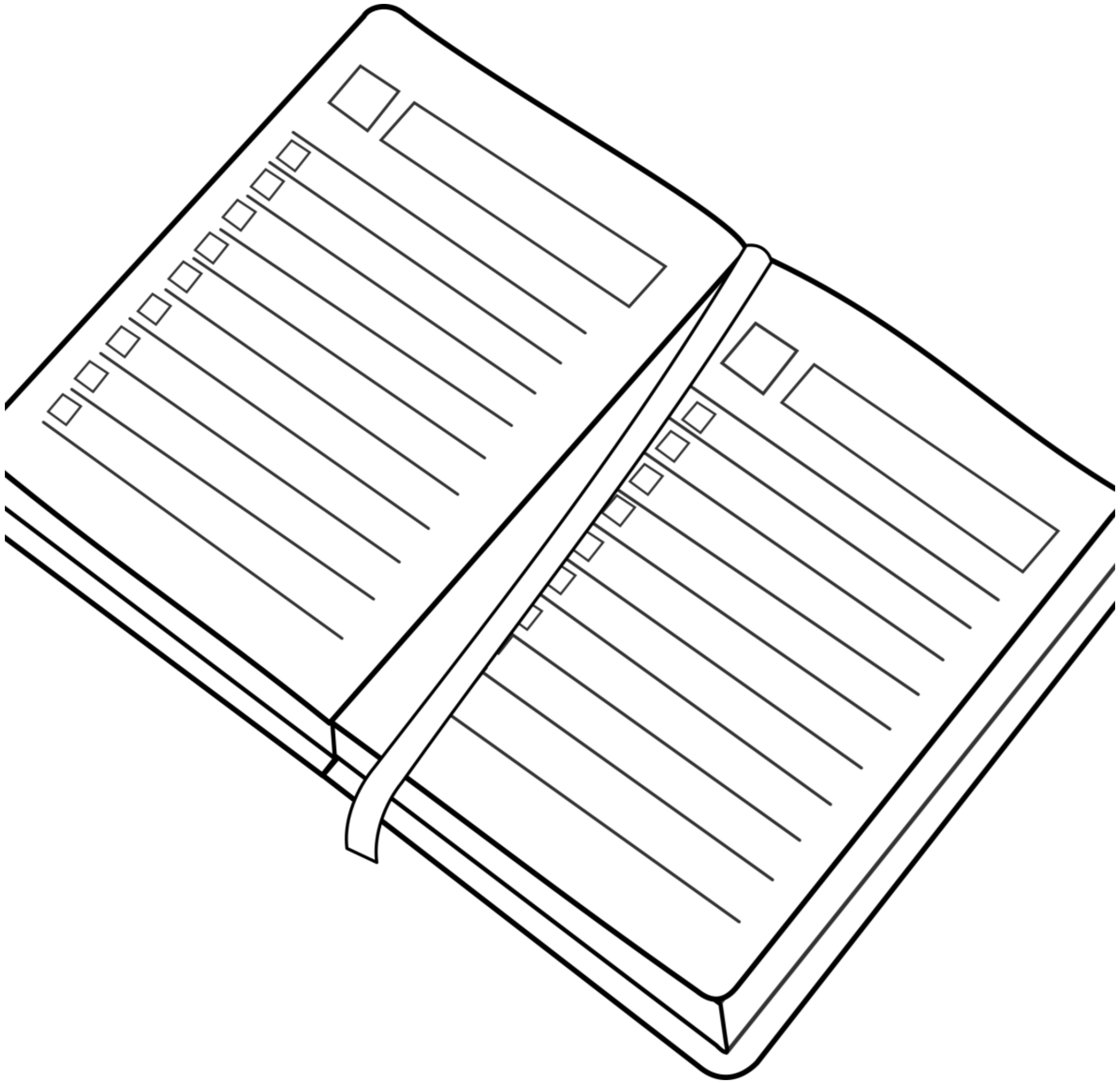
2026

Learning Objectives

- To understand about IPv4 Addressing and subnetting concept.
- Able to create subnets for any IPv4 given address.
- To understand the need of VLSM and able to create VLSM for any given IPv4 address.

Agenda

- The problem of Fixed Length Subnet Mask and “Wasted” addresses
- Variable Length Subnet Masks
- Example: VLSM Network Design
- Supernetting



Subnet to Meet Requirements

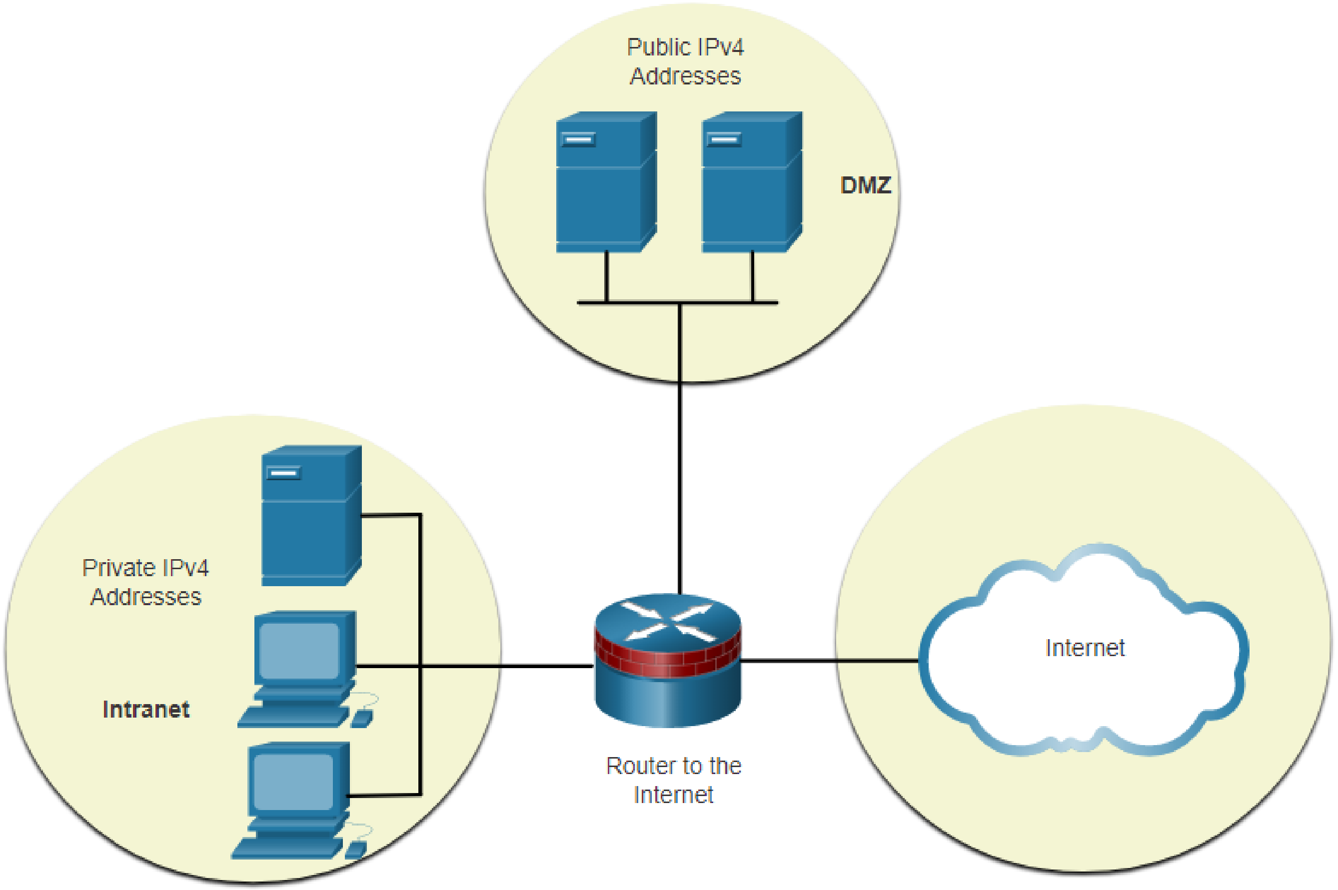
Subnet Private versus Public IPv4 Address Space

Enterprise networks will have an:

- Intranet - A company's internal network typically using private IPv4 addresses.
- DMZ – A companies internet facing servers. Devices in the DMZ use public IPv4 addresses.

- A company could use the 10.0.0.0/8 and subnet on the /16 or /24 network boundary.

- The DMZ devices would have to be configured with public IP addresses.



Subnetting: Minimise "Wasted" Addresses

- When planning subnets, generally want most efficient use of Network space
- Table below relates to subnetting a /24 network (e.g. 192.168.20.0/24).
  - Enough subnets?
  - Enough Hosts?

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.**n**hhhhhhh 11111111.11111111.11111111.**1**0000000	**2**	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nn**hhhhhh 11111111.11111111.11111111.**11**000000	**4**	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnn**hhhhh 11111111.11111111.11111111.**111**00000	**8**	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnn**hhhh 11111111.11111111.11111111.**1111**0000	**16**	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnn**hhh 11111111.11111111.11111111.**11111**000	**32**	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnnn**hh 11111111.11111111.11111111.**111111**00	**64**	2

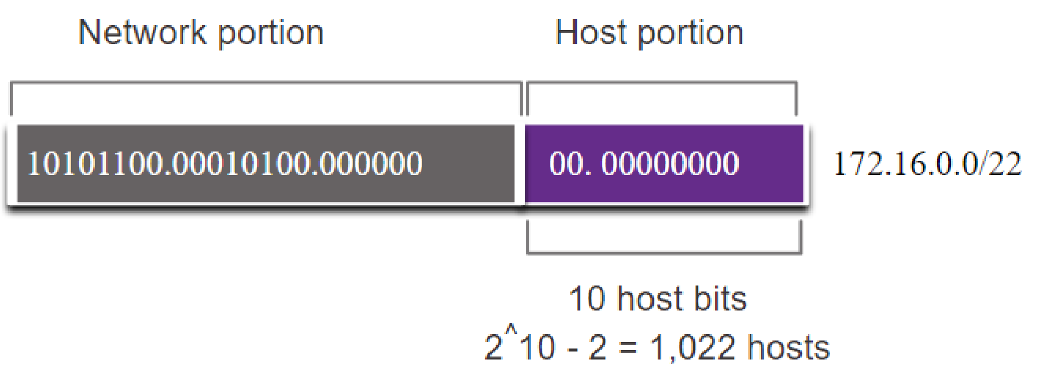
Create Subnets with a Slash 16 prefix

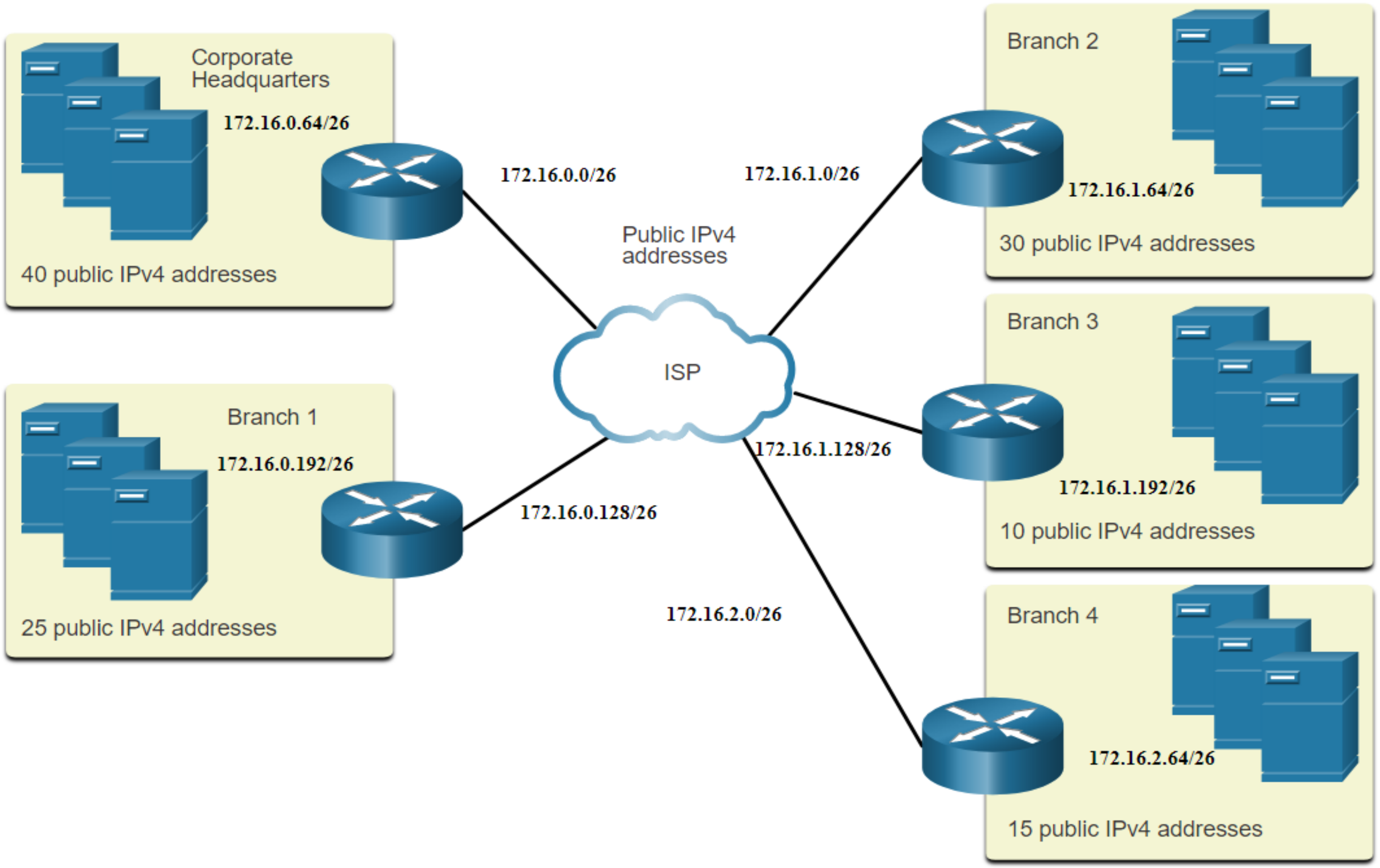
- The table highlights all the possible scenarios for subnetting a /16 prefix.

Prefix Length	Subnet Mask	Network Address (n = network, h = host)	# of subnets	# of hosts
/17	255.255.**128**.0	nnnnnnnn.nnnnnnnn.**n**hhhhhhh.hhhhhhhh 11111111.11111111.**1**0000000.00000000	**2**	32766
/18	255.255.**192**.0	nnnnnnnn.nnnnnnnn.**nn**hhhhhh.hhhhhhhh 11111111.11111111.**11**000000.00000000	**4**	16382
/19	255.255.**224**.0	nnnnnnnn.nnnnnnnn.**nnn**hhhhh.hhhhhhhh 11111111.11111111.**111**00000.00000000	**8**	8190
/20	255.255.**240**.0	nnnnnnnn.nnnnnnnn.**nnnn**hhhh.hhhhhhhh 11111111.11111111.**1111**0000.00000000	**16**	4094
/21	255.255.**248**.0	nnnnnnnn.nnnnnnnn.**nnnnn**hhh.hhhhhhhh 11111111.11111111.**11111**000.00000000	**32**	2046
/22	255.255.**252**.0	nnnnnnnn.nnnnnnnn.**nnnnnn**hh.hhhhhhhh 11111111.11111111.**111111**00.00000000	**64**	1022
/23	255.255.**254**.0	nnnnnnnn.nnnnnnnn.**nnnnnnn**h.hhhhhhhh 11111111.11111111.**1111111**0.00000000	**128**	510
/24	255.255.**255.0**	nnnnnnnn.nnnnnnnn.**nnnnnnnn.**hhhhhhhh 11111111.11111111.**11111111.**00000000	**256**	254
/25	255.255.**255.128**	nnnnnnnn.nnnnnnnn.**nnnnnnnn.n**hhhhhhh 11111111.11111111.**11111111.1**0000000	**512**	126
/26	255.255.**255.192**	nnnnnnnn.nnnnnnnn.**nnnnnnnn.nn**hhhhhh 11111111.11111111.**11111111.11**000000	**1024**	62
/27	255.255.**255.224**	nnnnnnnn.nnnnnnnn.**nnnnnnnn.nnn**hhhhh 11111111.11111111.**11111111.111**00000	**2048**	30
/28	255.255.**255.240**	nnnnnnnn.nnnnnnnn.**nnnnnnnn.nnnn**hhhh 11111111.11111111.**11111111.1111**0000	**4096**	14
/29	255.255.**255.248**	nnnnnnnn.nnnnnnnn.**nnnnnnnn.nnnnn**hhh 11111111.11111111.**11111111.11111**000	**8192**	6
/30	255.255.**255.252**	nnnnnnnn.nnnnnnnn.**nnnnnnnn.nnnnnn**hh 11111111.11111111.**11111111.111111**00	**16384**	2

Example: Efficient IPv4 Subnetting

- In this example, corporate headquarters has been allocated a public network address of 172.16.0.0/22 (10 host bits) by its ISP providing 1,022 host addresses.
- There are five sites and therefore five internet connections which means the organization requires 10 subnets with the largest subnet requires 40 addresses.
- It allocated 10 subnets with a /26 (i.e., 255.255.255.192) subnet mask.





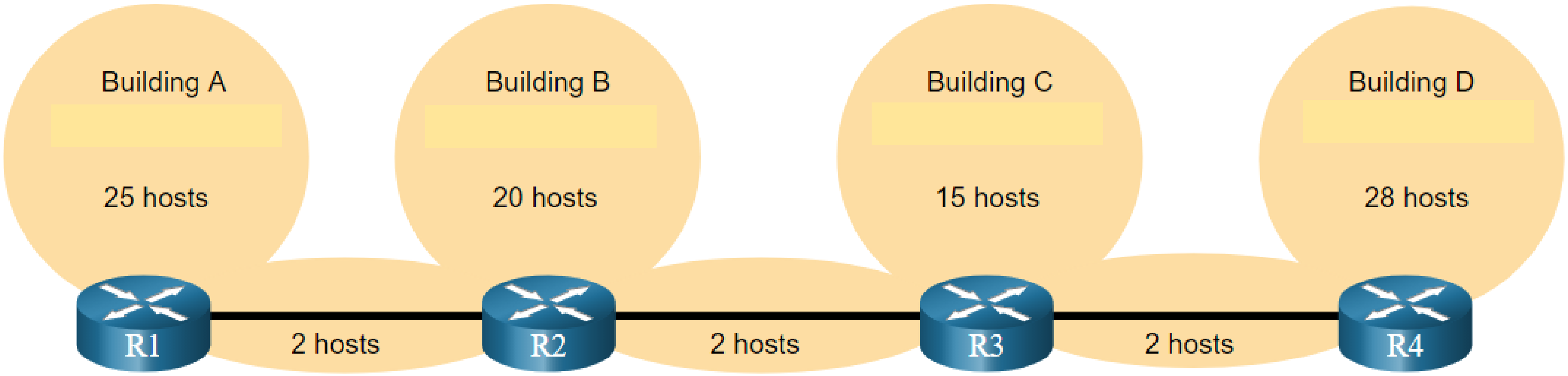
Basic IPv4 Subnetting Explanation – Video

Basic VSLM Explanation – Video

VSLM Explanation – Video

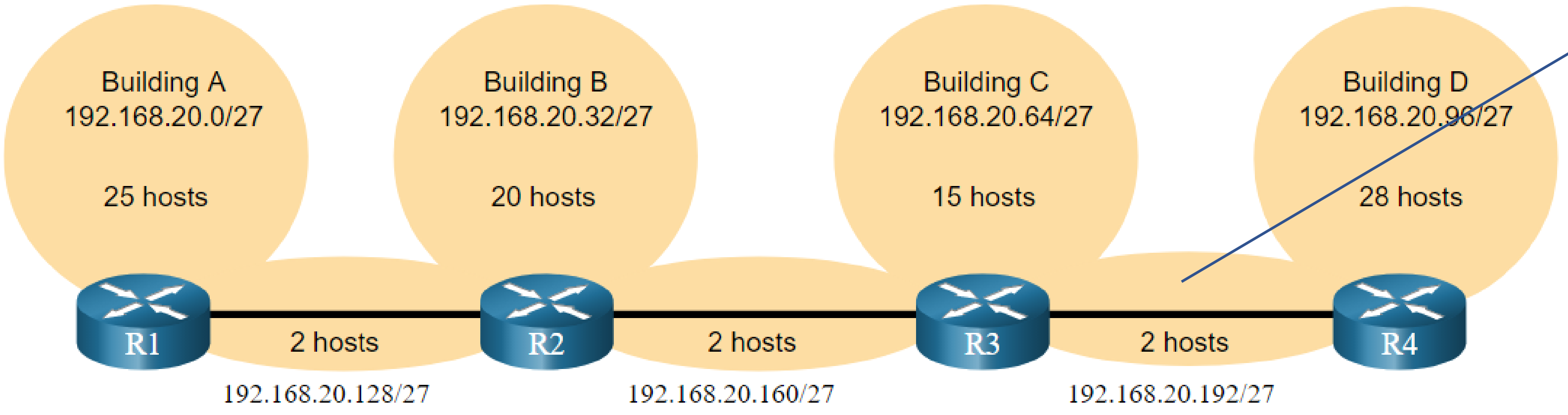
Subnetting Problem/Task

- Given **192.168.20.0/24, create 7 subnets, largest subnet: 28 hosts, smallest: 2 hosts**



Minimise “wasted IP” addresses

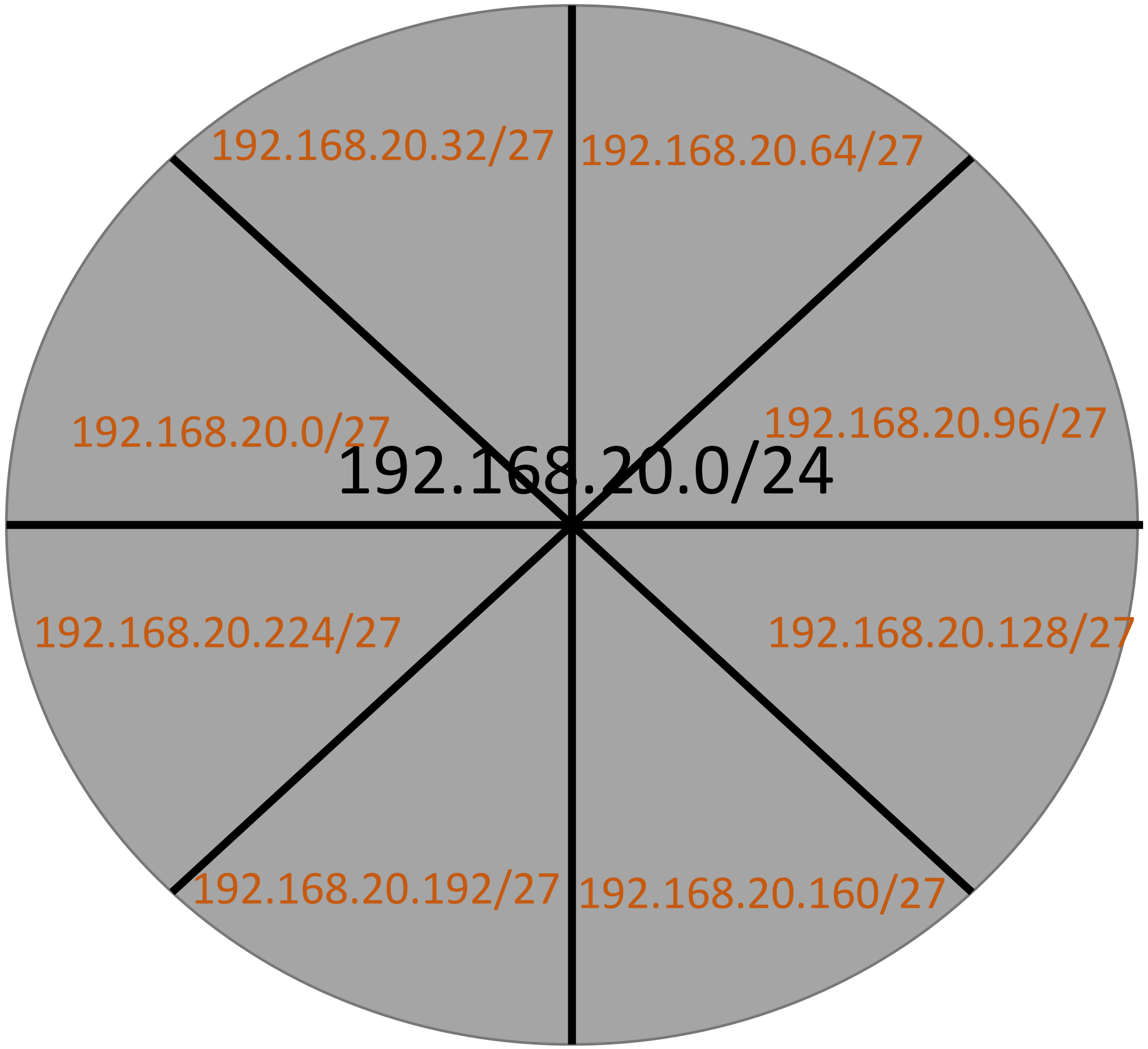
- Generally you wish to conserve IP addresses:
  - Given **192.168.20.0/24, create 7 subnets, largest subnet: 28 hosts, smallest: 2 hosts**
  - Using Fixed Length Subnet Masks, the following would result from subnetting **192.168.20.0/24** for the following topology:



Point to Point links between Routers only need 2 addresses. 28 “wasted” addresses

Visual Representation

192.168.20.0/24



VLSM

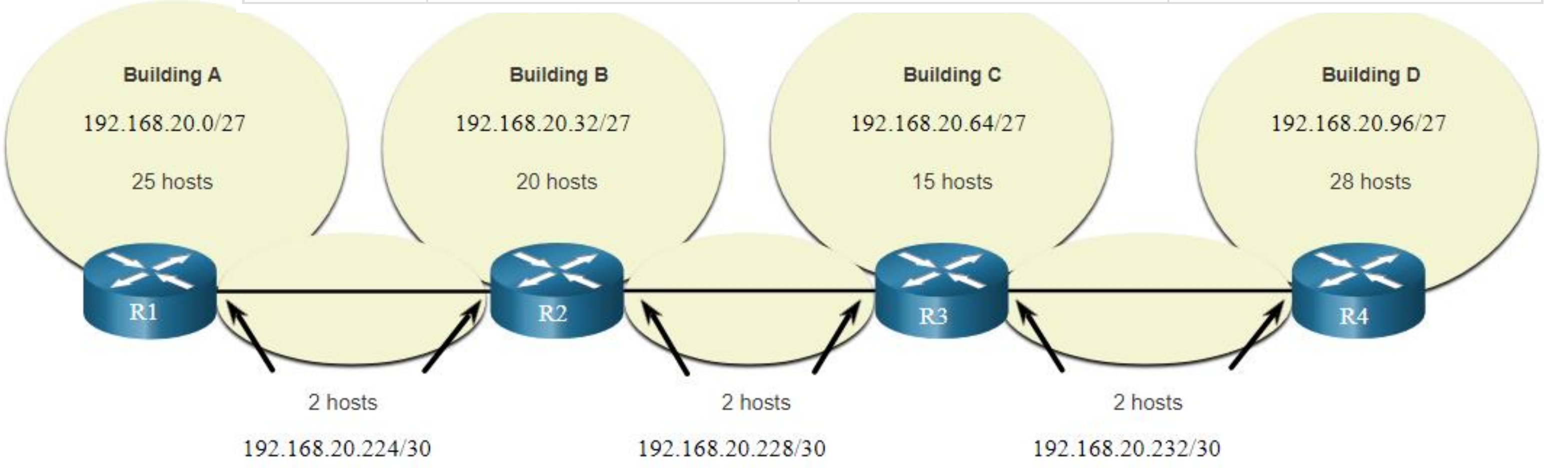
- VLSM was developed to avoid wasting addresses by enabling us to **subnet a subnet.**
- VLSM can be used to subnet a subnet
- In this case, divided the last subnet into eight /30 subnets.

Traditional Subnetting Creates Equal Sized Subnets

Subnets of Varying Sizes

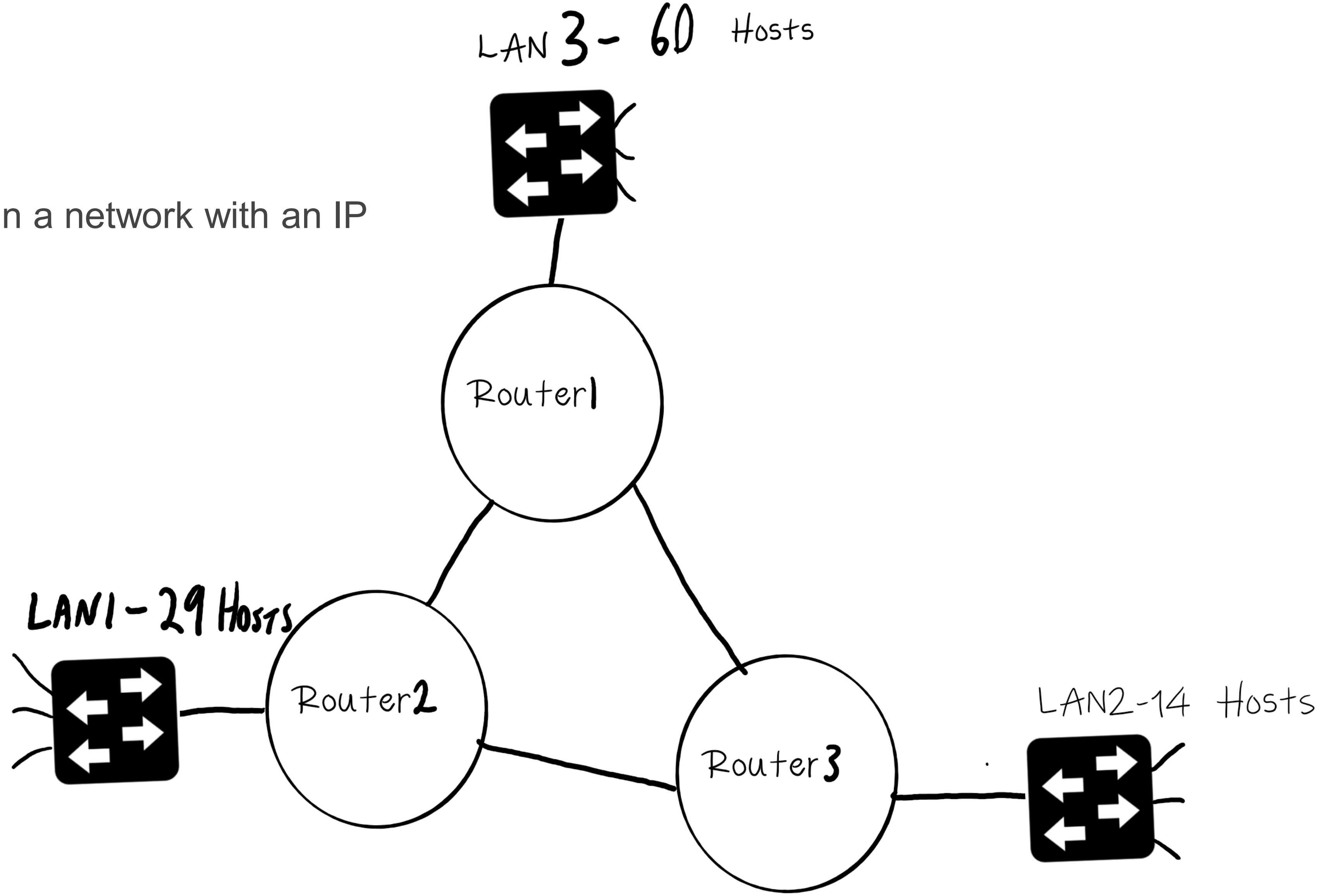
One subnet was further divided to create 8 smaller subnets of 2 hosts each.

	Network Address	Range of Host Addresses	Default Gateway Address
Building A	192.168.20.0/27	192.168.20.1/27 to 192.168.20.30/27	192.168.20.1/27
Building B	192.168.20.32/27	192.168.20.33/27 to 192.168.20.62/27	192.168.20.33/27
Building C	192.168.20.64/27	192.168.20.65/27 to 192.168.20.94/27	192.168.20.65/27
Building D	192.168.20.96/27	192.168.20.97/27 to 192.168.20.126/27	192.168.20.97/27
R1-R2	192.168.20.224/30	192.168.20.225/30 to 192.168.20.226/30	
R2-R3	192.168.20.228/30	192.168.20.229/30 to 192.168.20.230/30	
R3-R4	192.168.20.232/30	192.168.20.233/30 to 192.168.20.234/30	



Example: VLSM

Using the VLSM technique, design a network with an IP range of **192.168.4.0/24**.

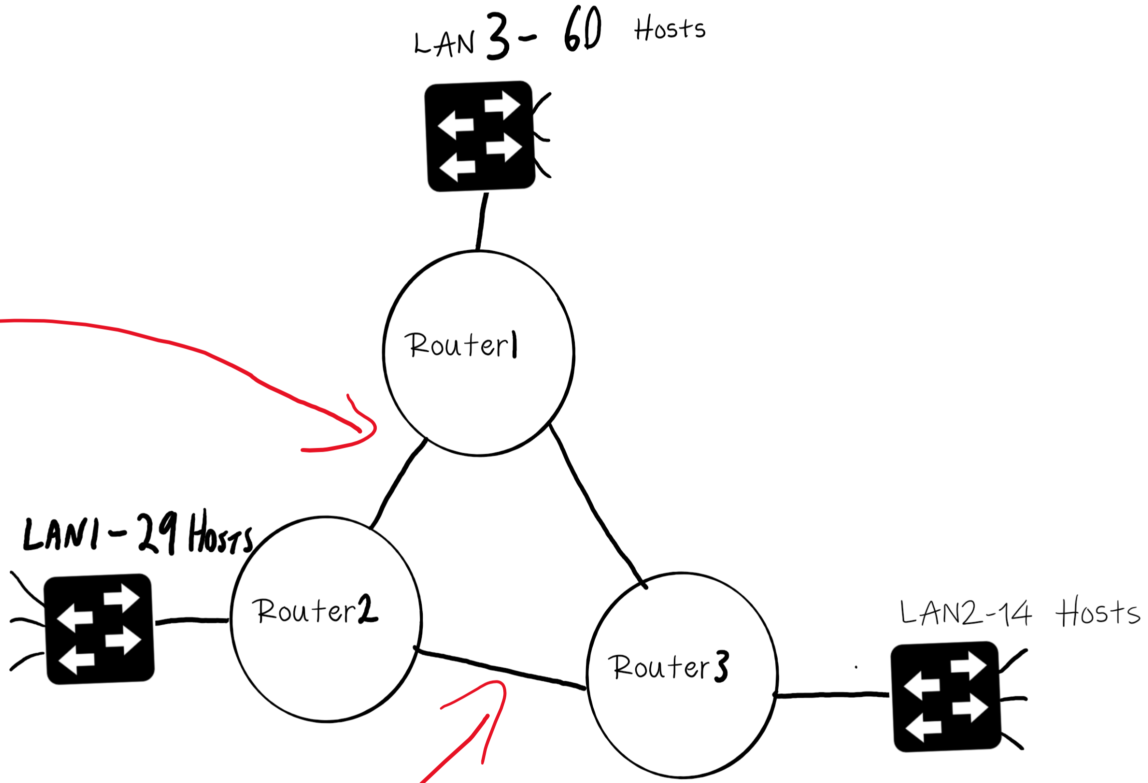


Example: Subnetting Reference Table

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.**n**hhhhhhh 11111111.11111111.11111111.**1**0000000	**2**	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nn**hhhhhh 11111111.11111111.11111111.**11**000000	**4**	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnn**hhhhh 11111111.11111111.11111111.**111**00000	**8**	30
/28	255.255.255.240	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnn**hhhh 11111111.11111111.11111111.**1111**0000	**16**	14
/29	255.255.255.248	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnn**hhh 11111111.11111111.11111111.**11111**000	**32**	6
/30	255.255.255.252	nnnnnnnn.nnnnnnnn.nnnnnnnn.**nnnnnn**hh 11111111.11111111.11111111.**111111**00	**64**	2

Example: Network Sizes

LAN	# Host
LAN 3	60
LAN 1	29
LAN 2	14
Link Router 1-2	2
Link Router 1-3	2
Link Router 2-3	2



1. Sort Networks, by max hosts, in descending order

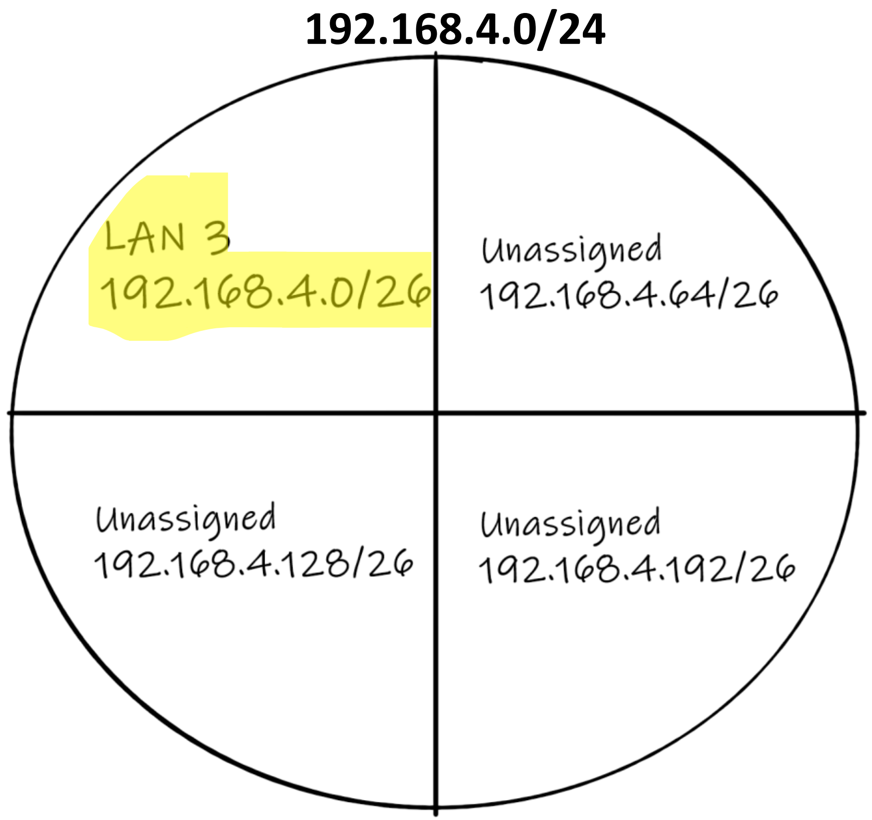
LAN	# Host
LAN 3	60
LAN 1	29
LAN 2	14
Link Router 1-2	2
Link Router 1-3	2
Link Router 2-3	2

2: Assign Network Address to Largest Network (LAN 3)

Prefix Length	Subnet Mask	Subnet Mask in Binary (n = network, h = host)	# of subnets	# of hosts
/25	255.255.255.128	nnnnnnnn.nnnnnnnn.nnnnnnnn.nhhhhhhh 11111111.11111111.11111111.10000000	2	126
/26	255.255.255.192	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnhhhhhh 11111111.11111111.11111111.11000000	4	62
/27	255.255.255.224	nnnnnnnn.nnnnnnnn.nnnnnnnn.nnnhhhhh	8	30

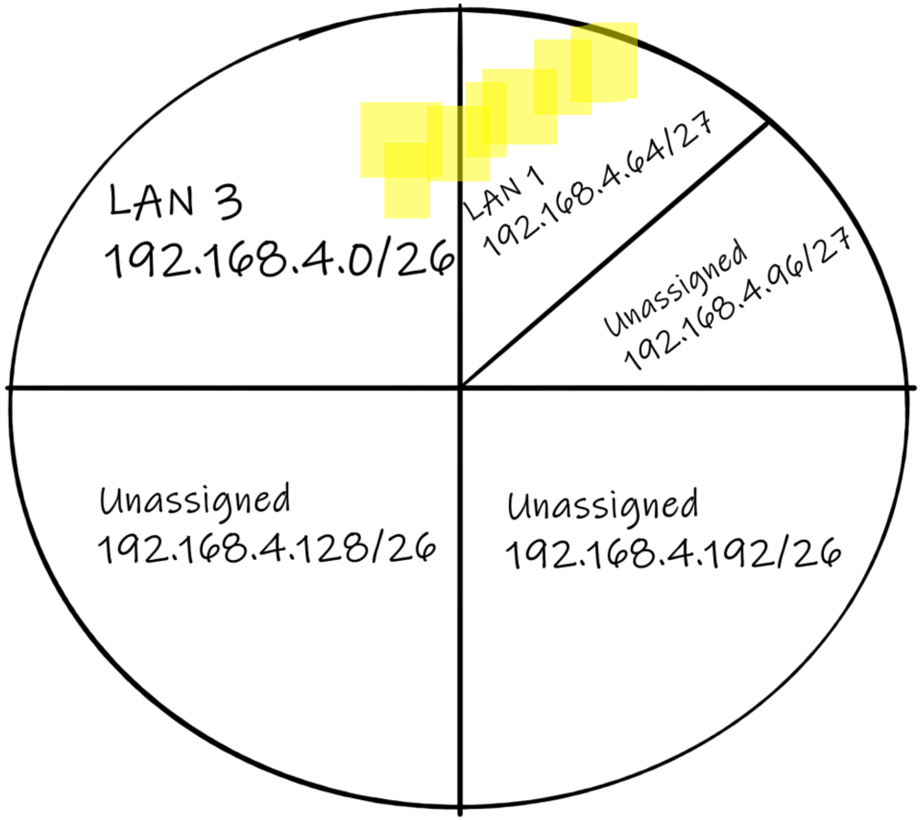
- We need 60 hosts (LAN 3)
- Refer to Subnetting Table, closest match that accommodates 60 hosts is /26 (64-2 = 62 useable addresses)

Network Address	Subnet Mask	Last Octet of Subnet Mask	Host Addresses (2^6-2)	Usable Host Range	Name of LAN
192.168.4.0	/26	.00000000	62	192.168.4.1–192.168.4.62	LAN 3
192.168.4.64	/26	.01000000	62		Unassigned
192.168.4.128	/26	.10000000	62		Unassigned
192.168.4.192	/26	.11000000	62		Unassigned



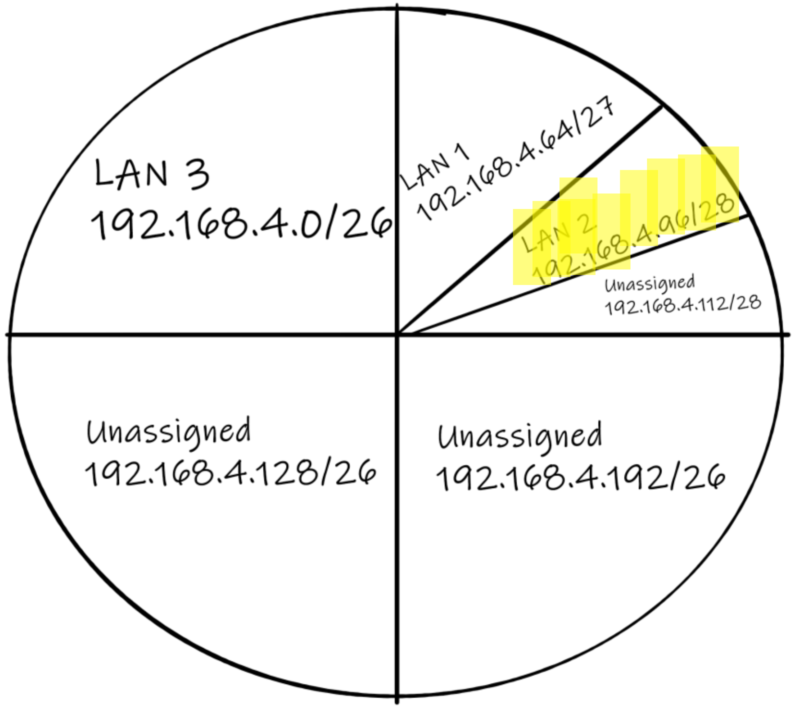
2. Second Largest Network (LAN 1)

- 29 Hosts (LAN 1)
- Refer to Subnet Table, closest match that accommodates **29** hosts is **/27** (32-2 = 30 useable addresses)
- Select the first unassigned large subnet in previous step and subdivide into two smaller subnets, **192.168.4.64/27** and **192.168.4.96/27**



Network ID	Subnet Mask	Host Addresses(2^5 -2)	Usable Host Range	Name of LAN
192.168.4.64	/27	30	192.168.4.65 – 192.168.4.94	LAN 1
192.168.4.96	/27	30		Unassigned

3. Third Largest Network



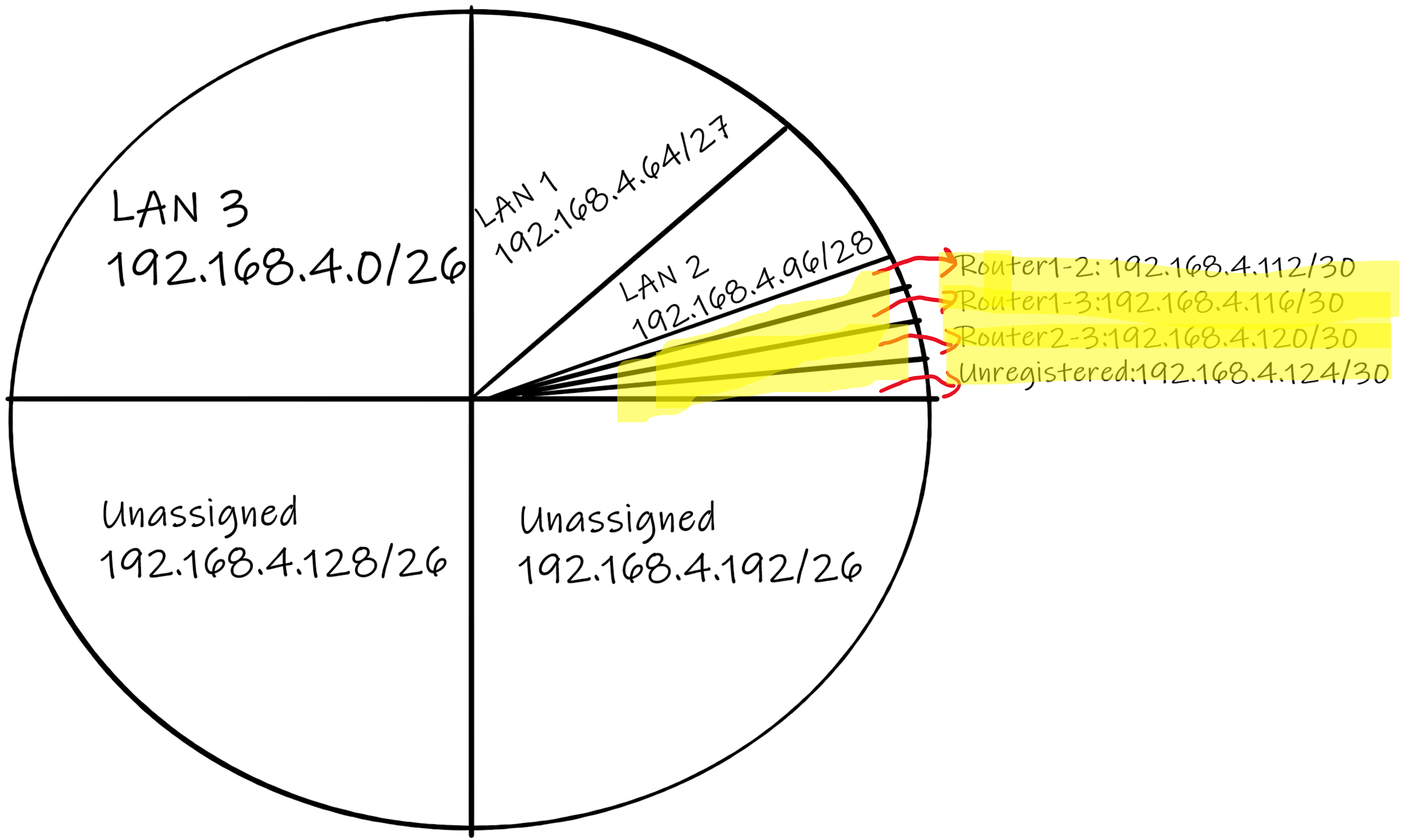
- 14 Hosts (LAN 2)
- Refer to Subnet Table, closest match that accommodates 14 hosts is /28 (16-2 = 14 useable addresses)
- Select the first unassigned large subnet in previous step and subdivide into two smaller subnets, 192.168.4.96/28 and 192.168.4.112/28

Network ID	Subnet Mask	Host Addresses (2^4 -2)	Usable Host Range	Name of LAN
192.168.4.96	/28	14	192.168.4.97–192.168.4.110	LAN 2
192.168.4.112	/28	14		Unassigned

4. Links

- Assign three smaller subnets for links between Routers1-2,1-3 and 2-3
- Just 2 Hosts/interfaces per link
- Refer to Subnet Table, closest match that accommodates 2 hosts is /30 (4-2 = 2 useable addresses)
- Select the first unassigned large subnet in previous step and subdivide into four smaller subnets, 192.168.4.112/30, 192.168.4.114/30….

Network ID	Subnet Mask	Host Addresses(2^2 -2)	Usable Host Range	Name of LAN
192.168.4.114	/30	2	192.168.4.113–192.168.4.114	Router1-2
192.168.4.116	/30	2	192.168.4.117–192.168.4.118	Router1-3
192.168.4.120	/30	2	192.168.4.121–192.168.4.122	Router2-3
192.168.4.124	/30	2		Unassigned


LAN 3
192.168.4.0/26
LAN 1
192.168.4.64/27
LAN 2
192.168.4.96/28
Router1-2: 192.168.4.112/30
Router1-3:192.168.4.116/30
Router2-3:192.168.4.120/30
Unregistered:192.168.4.124/30
Unassigned
192.168.4.128/26
Unassigned
192.168.4.192/26

Final Network Design

Network ID	Subnet Mask	Total Addresses(Inc. Network and Broadcast)	Usable Host Range	Name of LAN
192.168.4.0	/26	64	192.168.4.1–192.168.4.62	LAN 3
192.168.4.64	/27	32	192.168.4.65 – 192.168.4.94	LAN 1
192.168.4.96	/28	16	192.168.4.97– 192.168.4.110	LAN 2
192.168.4.112	/30	4	192.168.4.113–192.168.4.114	LINK Router1-2
192.168.4.116	/30	4	192.168.4.117–192.168.4.118	LINK Router1-3
192.168.4.120	/30	4	192.168.4.121–192.168.4.122	LINK Router2-3

Summary: VLSM Network Design Method

- Sort Subnets in descending order by Max number of hosts
- Select Largest Network and assign the most suitable network address by subnetting base network address.
- Assign address to next Largest Network (Subnet next unassigned network if possible)
- Repeat until all network address.

Fixed Length Subnet Masks vs Variable Length Subnet Masks

FLSM (Fixed Length Subnet Masks) Subnetting	VLSM (Variable Length Subnet Masks) Subnetting
Old-fashioned	Modern
Subnets are equal in size	Subnets are variable in size.
Subnets have an equal number of hosts	Subnets have a variable number of hosts
Supports both classful and classless routing protocols	Supports only classless routing protocols
Wastes more IP addresses	Wastes fewer IP addresses
Subnets use the same subnet mask	Subnets use different subnet masks
Simple configuration and administration	Complex configuration and administration

Source: https://www.comparitech.com/net-admin/variable-length-subnet-mask-vlsm-tutorial/

IPv4 Network Address Planning

IP network planning is crucial to develop a scalable solution to an enterprise network.

- To develop an IPv4 network wide addressing scheme, you need to know how many subnets are needed, how many hosts a particular subnet requires, what devices are part of the subnet, which parts of your network use private addresses, and which use public, and many other determining factors.

Examine the needs of an organization's network usage and how the subnets will be structured.

- Perform a network requirement study by looking at the entire network to determining how each area will be segmented.
- Determine how many subnets are needed and how many hosts per subnet.
- Determine DHCP address pools and Layer 2 VLAN pools.

Device Address Assignment

Within a network, there are different types of devices that require addresses:

- **End user clients** – Most use DHCP to reduce errors and burden on network support staff. IPv6 clients can obtain address information using DHCPv6 or SLAAC.
- **Servers and peripherals** – These should have a predictable static IP address.
- **Servers that are accessible from the internet** – Servers must have a public IPv4 address, most often accessed using NAT.
- **Intermediary devices** – Devices are assigned addresses for network management, monitoring, and security.
- **Gateway** – Routers and firewall devices are gateway for the hosts in that network.

When developing an IP addressing scheme, it is generally recommended that you have a set pattern of how addresses are allocated to each type of device.

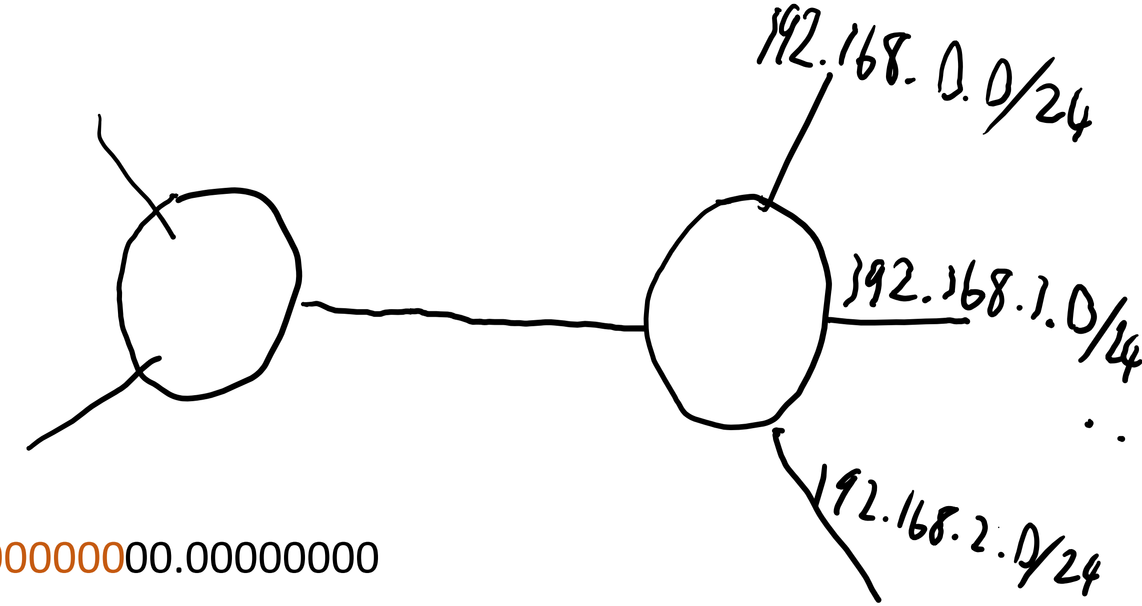
Supernetting

- Also known as route aggregation or CIDR
- Often used in "route summarisation"
- Routes to multiple networks/subnets with similar network prefixes are combined into a single routing entry
- Common in larger networks and on the internet to ensure efficient use of IP addresses and to reduce the size of routing tables.

Superneting and Subnetting

- At the local level, an organisation might use subnetting to segment its internal networks(like the example above)
- At a higher hierarchical level, when advertising routes to external entities or upstream providers, supernetting can be used. Instead of advertising each individual subnet, the organization can advertise a **summarized or aggregated route**.
- This reduces the number of routes that external entities need to know about, which in turn reduces the size of their routing tables.
- This combination allows for detailed network organization internally (through subnetting) while presenting a simplified view to the outside world (through supernetting).

Supernetting example



Combine the following into one summarised route

192.168.0.0/24	11000000.10101000.00000000.00000000
192.168.1.0/24	11000000.10101000.00000001.00000000
192.168.2.0/24	11000000.10101000.00000010.00000000
192.168.3.0/24	11000000.10101000.00000011.00000000

11000000.10101000.00000000.00000000 = **192.168.0.0**

11111111.11111111.11111100.00000000 = **255.255.252.0**

1. Write Network IP addresses in binary
2. Find matching bits from left to right
3. Keep the matching numbers and add the remaining zeros, this is your SuperNet (Summarised network address
4. For the subnet mask. Put “1s” in the matching networking part